

Build a To-Do App

Next.js · TypeScript · Node.js · PostgreSQL · Zustand: a beginner's study guide

This guide walks you through a small but **real** full-stack app: a page where you add, tick and delete tasks, saved in a PostgreSQL database. It's about 150 lines of code in 7 files, small enough to memorize and big enough to teach the things LeetCode can't:

- how a browser talks to a server (**HTTP, APIs, JSON**)
- how data is stored and queried (**databases, SQL**)
- how a project is organised (**folders, config, secrets**)
- how a UI keeps track of data (**state, Zustand**)

Every file is shown with a **line-by-line explanation** table: code on the left, meaning on the right. All commands are for **Windows PowerShell**.

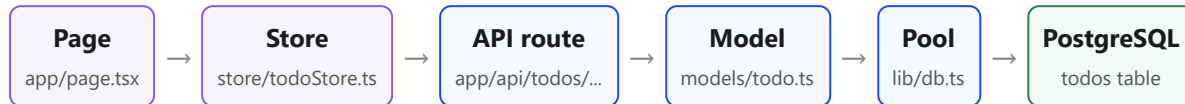
Part 1	Understand before you type: the big picture, key words, folders, who creates which file
Part 2	Set up PostgreSQL: install, psql, create the database and table
Part 3	Create the Next.js project with one command, add libraries
Part 4	Write the code, back to front: database → model → API → store → page
Part 5	Compare with Express + Mongoose
Part 6	Memorize and practice, plus troubleshooting

Part 1: Understand before you type

Ten minutes here saves hours of confusion later.

The big picture

When you click **Add**, your task travels through these pieces and back:



Purple = runs in the **browser**. Blue = runs on the **server** (Node.js). Green = the **database**.

IN PLAIN ENGLISH

Think of a restaurant:

Restaurant	Our app	Job
Dining room	Page	What the customer sees and clicks.
Order pad	Store (Zustand)	Remembers the current list; sends orders to the kitchen.
Waiter at the kitchen window	API route	Takes an order (request), checks it makes sense, returns the dish (response).
Cook	Model	Knows the recipes, i.e. the SQL for each job.
Phone lines to the pantry	Pool	A few always-open lines to the database, shared by everyone.
Pantry	PostgreSQL	Where everything is stored permanently.

Words you'll see

Word	Meaning
Node.js	Runs JavaScript outside the browser, on your computer or a server. Next.js runs on it.
npm	Node's package manager. Downloads libraries into <code>node_modules/</code> and lists them in <code>package.json</code> .
Next.js	A framework that serves your pages and your API from one project. It replaces Express here.
React	The library for building the UI out of components (functions that return HTML-like JSX).
TypeScript	JavaScript plus types . Catches mistakes (like a typo in a field name) before you run the code.
State	Data that, when it changes, makes the screen redraw. <code>useState</code> (one component) or a store (shared).
Hook	A React function starting with <code>use...</code> (<code>useState</code> , <code>useEffect</code> , <code>useTodoStore</code>).
Zustand	A tiny library for shared state : one store any component can read and update.
API / endpoint	A URL that returns data (JSON) instead of a page, e.g. <code>/api/todos</code> .
HTTP method	The verb of a request: GET read, POST create, PATCH update, DELETE remove.

Status code	A number in each reply: 200 OK, 201 Created, 204 No Content, 400 Bad Request, 404 Not Found, 500 Server Error.
JSON	Text format for data: <code>{"id": 1, "text": "Buy milk", "done": false}</code> .
PostgreSQL	A relational database: data lives in tables with fixed columns ; each item is a row .
SQL	The language for talking to the database: <code>SELECT</code> , <code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code> .
psql	PostgreSQL's terminal app for typing SQL. Like <code>mongosh</code> for MongoDB.
pg	The npm library your Node.js code uses to talk to PostgreSQL.
Connection pool	A few database connections kept open and shared, so each request doesn't open a new one.
Environment variable	A setting kept outside the code (in <code>.env.local</code>), e.g. the database password.

The folders: is this the standard design?

project structure

```

todo-app/
├─ app/                ← REQUIRED by Next.js: pages and API routes
│  ├─ layout.tsx       ← frame around every page
│  ├─ page.tsx         ← the page at /
│  └─ api/todos/
│     ├─ route.ts       ← API: GET, POST /api/todos
│     └─ [id]/route.ts  ← API: PATCH, DELETE /api/todos/7
├─ lib/db.ts           ← convention: shared helpers (database pool)
├─ models/todo.ts      ← convention: all the SQL for todos
├─ store/todoStore.ts  ← convention: Zustand store
├─ schema.sql          ← the table definition (run once with psql)
├─ .env.local          ← secrets: database address + password
└─ package.json, tsconfig.json, next.config.ts ... ← config (generated)

```

Folder	Required?	Why it's there
app/	Yes	Next.js's App Router turns this folder into URLs. <code>page.tsx</code> = a page, <code>route.ts</code> = an API endpoint, <code>[id]</code> = a changing part of the URL. Those file names are fixed.
lib/	No	Very common convention for helpers and setup code. You could call it <code>utils/</code> ; Next.js doesn't care.
models/	No	Our choice: puts all SQL in one place, like Mongoose models. Keeps API routes short.
store/	No	Our choice: where Zustand stores live.

IN PLAIN ENGLISH

Next.js only cares about `app/` (and `public/` for images, which we don't use). Everything else is a tidy habit. Other projects often add `components/` , `hooks/` , `types/` , or put everything inside `src/` . All of those are normal.

Who creates which file?

Many files appear "by magic". Here is where each one comes from and whether you touch it:

File / folder	Created by	You edit it?	What it is
package.json	create-next-app	Rarely	Project name, scripts (<code>npm run dev</code>), list of libraries.
package-lock.json	npm install	Never	Exact versions of every library, so installs are repeatable.
node_modules/	npm install	Never	The downloaded libraries (Next.js, React, pg...). Huge; never commit it.
tsconfig.json	create-next-app	No	TypeScript settings. (If missing, <code>next dev</code> creates it. <code>npm init -y</code> does not ; that only makes <code>package.json</code> .)
next.config.ts	create-next-app	Later	Next.js settings. Empty = defaults.
next-env.d.ts	Next.js	Never	Tells TypeScript about Next.js's types.
.git/	create-next-app (if Git is installed)	Never directly	Git's history of your project. Use <code>git status</code> , <code>git add</code> and <code>git commit</code> instead of touching it.
.next/	create-next-app, then <code>next dev</code> / <code>next build</code>	Never	Next.js's output and cache : your code compiled for the browser and server. It is <i>not</i> the npm package (that's <code>node_modules/next</code>); it's what the package <i>produces</i> when it runs. Safe to delete; it's rebuilt.
tsconfig.tsbuildinfo	TypeScript	Never	Cache that makes type-checking faster.
.gitignore	create-next-app	Rarely	Files Git should never save (<code>node_modules</code> , <code>.next</code> , <code>.env*</code>).
AGENTS.md , CLAUDE.md , README.md	create-next-app	Optional	Notes for humans and AI coding assistants. Not used by the app.
app/layout.tsx , app/page.tsx	create-next-app	Yes	Generated as a "Hello world", then you replace them.
.env.local , schema.sql , lib/ , models/ , store/ , app/api/	You	Yes	Your actual app.
guide/	-	-	Not part of the app. <code>build-guide.mjs</code> and <code>explanations.mjs</code> only generate this PDF. You can delete the folder and the app still works.

Part 2: Set up PostgreSQL

Install the database, meet psql, create the database and the table.

1 Install Node.js

Download the LTS version from nodejs.org and install it. Check it in PowerShell:

```
PowerShell
```

```
node -v  
npm -v
```

2 Install PostgreSQL

PostgreSQL is a program that runs in the background (a Windows **service**) and waits for connections on **port 5432**. Pick one way to install it:

Option A: one command (what we did)

Choose a password first: letters and numbers only, no spaces. Then run:

```
PowerShell
```

```
winget install --id PostgreSQL.PostgreSQL.17 --source winget --override "--mode unattended --  
unattendedmodeui none --superpassword YOUR_PASSWORD --serverport 5432"
```

Part	Meaning
<code>winget install --id PostgreSQL.PostgreSQL.17</code>	Use Windows' package manager to install PostgreSQL version 17.
<code>--override "..."</code>	Pass these options straight to the PostgreSQL installer:
<code>--mode unattended --unattendedmodeui none</code>	Install without asking any questions.
<code>--superpassword YOUR_PASSWORD</code>	Password for the main database user, called <code>postgres</code> .
<code>--serverport 5432</code>	The standard PostgreSQL port.

WATCH OUT

Windows will show an **administrator (UAC) prompt**; click Yes. It may hide behind other windows (look for a flashing shield on the taskbar). The download plus install took us about 15 minutes. **Write the password down**; you need it in step 10.

Option B: the normal installer

Download the Windows installer from postgresql.org/download/windows and click Next through it. Set a password when asked, keep port **5432**, and untick "Stack Builder" at the end.

Check that it's running

```
PowerShell
```

```
sc.exe query postgresql-x64-17
```

Look for `STATE : 4 RUNNING`. It starts automatically every time Windows starts. The installer also adds **pgAdmin 4**, a point-and-click app for browsing your data (like MongoDB Compass).

3 Meet psql

IN PLAIN ENGLISH

psql is a terminal "chat window" with your database: you type SQL or commands, it answers. It's the PostgreSQL version of `mongosh`. You use it to set things up and peek at data; your **app** doesn't use psql, it uses the `pg` library.

psql lives in `C:\Program Files\PostgreSQL\17\bin`. Windows doesn't know that folder yet, so typing `psql` gives "not recognized". Fix it once by adding the folder to your **PATH**:

1. Press Start, type **environment variables**, open "Edit the system environment variables".
2. Click **Environment Variables...**, select **Path** under "User variables", then **Edit**, then **New**.
3. Paste `C:\Program Files\PostgreSQL\17\bin`, click OK three times, then **open a new PowerShell window**.

(Without PATH, you can always type the full path: `& "C:\Program Files\PostgreSQL\17\bin\psql.exe"` ...)

The flags you'll use:

Flag	Meaning	Our value
<code>-U</code>	User to log in as	<code>postgres</code>
<code>-h</code>	Host (which computer)	<code>localhost</code> (this one)
<code>-d</code>	Database to open	<code>todo_app</code>
<code>-c</code>	Run one command and exit	<code>"SELECT * FROM todos;"</code>
<code>-f</code>	Run all commands in a file	<code>schema.sql</code>

TIP

psql asks for the password every time. To skip that in the current window only, run `$env:PGPASSWORD = "YOUR_PASSWORD"` first. It's forgotten when you close the window.

4 Create the database

One PostgreSQL server can hold many databases (one per project). Create ours:

PowerShell

```
psql -U postgres -h localhost -c "CREATE DATABASE todo_app;"
```

It answers `CREATE DATABASE`. Do this once; running it again says "already exists", which is harmless.

5 Create the table

Unlike MongoDB, PostgreSQL needs to know the **shape** of your data before you store any: a table with named, typed columns. Save this as `schema.sql` in your project folder (you'll create the folder in step 6, so you can also come back to this step):

schema.sql	
Code	What it means
-- Run this once to create the table.	A comment. In SQL, comments start with <code>--</code> . The database ignores it.
CREATE TABLE IF NOT EXISTS todos (	Make a new table called <code>todos</code> . <code>IF NOT EXISTS</code> means: if it's already there, do nothing instead of crashing.
id SERIAL PRIMARY KEY,	Column 1: <code>id</code> . <code>SERIAL</code> = Postgres fills in 1, 2, 3... for you. <code>PRIMARY KEY</code> = no two rows can share an id. (Like Mongo's <code>_id</code> , but a simple number.)
text TEXT NOT NULL,	Column 2: <code>text</code> , the task name. <code>TEXT</code> = any length string. <code>NOT NULL</code> = it must always have a value.
done BOOLEAN NOT NULL DEFAULT FALSE	Column 3: <code>done</code> , true or false. If you don't give one, it starts as <code>FALSE</code> .
);	Close the column list. The <code>;</code> ends the SQL command.

Run the file against the `todo_app` database, then check the result with `\d` (describe):

PowerShell
<pre>psql -U postgres -h localhost -d todo_app -f schema.sql psql -U postgres -h localhost -d todo_app -c "\d todos"</pre>
expected output
<pre>Column Type Nullable Default -----+-----+-----+----- id integer not null nextval('todos_id_seq'::regclass) text text not null done boolean not null false Indexes: "todos_pkey" PRIMARY KEY, btree (id)</pre>

Handy psql commands (inside `psql -U postgres -h localhost -d todo_app`)

Type this	It does	Mongo shell equivalent
<code>\l</code>	List databases	<code>show dbs</code>
<code>\c todo_app</code>	Switch database	<code>use todo_app</code>
<code>\dt</code>	List tables	<code>show collections</code>
<code>\d todos</code>	Show a table's columns	(no schema in Mongo)
<code>SELECT * FROM todos;</code>	Show all rows (don't forget the <code>;</code>)	<code>db.todos.find()</code>
<code>\q</code>	Quit	<code>exit</code>

Part 3: Create the Next.js project

One command makes the skeleton; two more add our libraries.

6 Generate the project (boilerplate)

Go to the folder where you keep projects, then run:

PowerShell

```
npx create-next-app@latest todo-app --ts --app --empty --no-tailwind --no-eslint --no-src-dir --no-react-compiler --import-alias "@/*" --use-npm
cd todo-app
```

Part	Meaning
<code>npx create-next-app@latest todo-app</code>	Download and run Next.js's official project generator; make a folder <code>todo-app</code> .
<code>--ts</code>	Use TypeScript.
<code>--app</code>	Use the App Router (the <code>app/</code> folder).
<code>--empty</code>	Minimal starter: just a "Hello world!" page, no demo images or CSS.
<code>--no-tailwind --no-eslint --no-react-compiler</code>	Skip extra tools, to keep it simple for learning.
<code>--no-src-dir</code>	Put <code>app/</code> at the top level, not inside <code>src/</code> .
<code>--import-alias "@/*"</code>	Make <code>@/</code> mean "project root" in imports.
<code>--use-npm</code>	Use npm (not yarn/pnpm/bun).

TIP

Prefer questions? Run just `npx create-next-app@latest todo-app` and pick custom settings that match the flags above: TypeScript **yes**, App Router **yes**, Tailwind / ESLint / `src/` **no**, import alias `@/*`.

IN PLAIN ENGLISH

Know Vite? This is the same idea as `npm create vite@latest my-app -- --template react-ts`: one command makes a ready-to-run starter. The difference is that Vite gives you a browser-only React app (you'd still need a separate backend like Express), while Next.js gives you pages *and* API routes in one project.

What it creates (this project was made with exactly this command):

created by create-next-app

```
todo-app/
├ .git/                ← a Git repository with one commit (if Git is installed)
├ .next/               ← generated route types (more appears when you run the app)
├ app/
│   ├── layout.tsx     ← you'll replace this
│   └── page.tsx       ← "Hello world!", you'll replace this
├ node_modules/       ← next, react, react-dom, typescript, @types/...
├ .gitignore
├ AGENTS.md, CLAUDE.md, README.md
├ next-env.d.ts
├ next.config.ts
├ package.json
├ package-lock.json
└ tsconfig.json
```

It also runs `npm install` for you and, if Git is installed, makes a first commit called "Initial commit from Create Next App". From then on, `git status` shows exactly which files **you** changed or added.

7 Add our three libraries

PowerShell

```
npm install pg zustand
npm install -D @types/pg
```

Package	What for
pg	Lets Node.js talk to PostgreSQL.
zustand	Shared state in the browser.
@types/pg	Type descriptions for <code>pg</code> , so TypeScript understands it. <code>-D</code> = a dev -dependency: only needed while coding, not to run the app.

IN PLAIN ENGLISH

Why isn't the install command long anymore? Without create-next-app you'd have to install everything by hand: `npm install next react react-dom pg zustand` plus `npm install -D typescript @types/node @types/react @types/react-dom @types/pg`. Every `@types/...` package teaches TypeScript about one JavaScript library. create-next-app already installed all of them except `@types/pg`, so that's all you add.

Some libraries ship their own types and need no `@types` package. `zustand` is one; `pg` isn't.

These commands change `package.json` (new lines under `dependencies` / `devDependencies`), `package-lock.json` and `node_modules/`.

8 Run it once

PowerShell

```
npm run dev
```

Open **http://localhost:3000** and you'll see "Hello world!". Next.js just filled the `.next/` folder with the compiled app. Leave it running: it reloads automatically as you save files. Press **Ctrl+C** to stop it.

9 Understand the generated config (no changes needed)

You won't edit these, but you should know what they say.

tsconfig.json	
Code	What it means
{	This file is one JSON object of TypeScript settings. create-next-app wrote it for you; you never need to change it.
"compilerOptions": {	Everything inside this block tells the TypeScript checker how to behave.
"target": "ES2017",	Which JavaScript version to aim for. Next.js handles old browsers itself, so this rarely matters.
"lib": ["dom", "dom.iterable", "esnext"],	Which built-in type definitions to load: browser things (<code>document</code> , <code>fetch</code>) and modern JavaScript.
"allowJs": true,	Plain <code>.js</code> files are allowed in the project too.
"skipLibCheck": true,	Don't type-check the insides of downloaded libraries. Faster, and avoids errors that aren't yours.
"strict": true,	Turn on all the strict safety checks. This is the whole point of TypeScript: catching mistakes like using something that might be <code>undefined</code> .
"noEmit": true,	TypeScript only checks ; it doesn't produce any files. Next.js does the real compiling.
"esModuleInterop": true,	Lets <code>import</code> work smoothly with older libraries (like <code>pg</code>).
"module": "esnext",	Use modern <code>import</code> / <code>export</code> syntax.
"moduleResolution": "bundler",	Find imported files the way modern bundlers (like Next.js) do.
"resolveJsonModule": true,	You may <code>import</code> a <code>.json</code> file.
"isolatedModules": true,	Every file must make sense on its own. Next.js's fast compiler needs this.
"jsx": "react-jsx",	How to treat the HTML-looking <code><div></code> code in <code>.tsx</code> files. You don't need to import React in every file.
"incremental": true,	Remember the last check so the next one is faster. That's what the <code>tsconfig.tsbuildinfo</code> cache file is.
"plugins": [{ "name": "next" }],	Turns on the Next.js helper inside your editor (better autocomplete and warnings).

<pre>"paths": { "@/*": ["./*"] },</pre>	A shortcut: <code>@/</code> means "the project's root folder". So <code>import { pool } from "@/lib/db"</code> works from any folder, with no <code>../../../../</code>. The <code>}</code> after it closes <code>compilerOptions</code>.
<pre>"include": ["next-env.d.ts", "**/*.ts", "**/*.tsx", ".next/types/**/*.ts", ".next/dev/types/**/*.ts", "**/*.mts"],</pre>	Which files to check: every <code>.ts</code> / <code>.tsx</code> file, plus types that Next.js generates inside <code>.next/</code>.
<pre>"exclude": ["node_modules"] }</pre>	Never check <code>node_modules</code> (the downloaded libraries).

next.config.ts	
Code	What it means
<pre>import type { NextConfig } from "next";</pre>	Bring in the <i>type</i> <code>NextConfig</code>, so your editor can autocomplete valid settings. (<code>import type</code> = only used for checking, removed when the app runs.)
<pre>const nextConfig: NextConfig = { /* config options here */ };</pre>	The settings object. It's empty, which means "use all the defaults". This is where you'd later add things like image domains or redirects.
<pre>export default nextConfig;</pre>	Hand the settings to Next.js. It reads this file when <code>npm run dev</code> starts.

Part 4: Write the code

Back to front: secret → database connection → model → API → store → page.

10 Store the database address in .env.local

Create a file named exactly `.env.local` in the project root. Use the password from step 2:

.env.local	
Code	What it means
<pre>DATABASE_URL=postgres://postgres:YOUR_PASSWORD@localhost:5432/todo_app</pre>	One setting, written as an address: <pre>postgres://USER:PASSWORD@HOST:PORT/DATABASE</pre> Read it as: log in as user <code>postgres</code> with your password, on this computer (<code>localhost</code>), port <code>5432</code> , database <code>todo_app</code> . Code reads it with <code>process.env.DATABASE_URL</code> .

IN PLAIN ENGLISH

Next.js loads `.env.local` automatically when it starts, and only server code can read it, so the password never reaches the browser. `.gitignore` already lists `.env*` , so it won't be uploaded to GitHub. **If you change this file, restart `npm run dev` .**

11 Connect to the database: lib/db.ts

Create a folder `lib` and this file:

lib/db.ts	
Code	What it means
<pre>import { Pool } from "pg";</pre>	Bring in <code>Pool</code> from the <code>pg</code> library (the Node.js driver for PostgreSQL).
<pre>// A Pool keeps a few database connections open and lends one to each query. // DATABASE_URL comes from .env.local (Next.js loads it automatically, server-side only).</pre>	Comments. Read the pool deep-dive below for the full story.
<pre>function createPool() { return new Pool({ connectionString: process.env.DATABASE_URL }); }</pre>	A small function that builds a new pool. The only setting is the database address from <code>.env.local</code> . Everything else uses <code>pg</code> 's defaults (up to 10 connections).
<pre>// In development, Next.js re-runs this file every time you save. // Parking the pool on globalThis lets us reuse it instead of opening new connections each time.</pre>	Comments explaining the next three lines: a fix for a development-only problem.
<pre>const globalForDb = globalThis as unknown as { pool?: Pool };</pre>	<code>globalThis</code> is an object that survives when Next.js reloads your files. This line tells TypeScript: "treat it as an object that <i>might</i> have a <code>pool</code> on it". (<code>as unknown as ...</code> is a TypeScript cast, i.e. "trust me on the type".)

<pre>export const pool = globalForDb.pool ?? createPool();</pre>	The pool everyone uses. <code>??</code> means "if the left side is empty, use the right side". So: reuse the saved pool if one exists, otherwise create a new one.
<pre>if (process.env.NODE_ENV !== "production") { globalForDb.pool = pool; }</pre>	While developing (not in production), save the pool on <code>globalThis</code> , so after the next reload the line above finds it and reuses it.

Deep dive: what is a connection pool?

IN PLAIN ENGLISH

Opening a connection to PostgreSQL is slow: the network handshake plus logging in with a password. And Postgres only accepts about **100** connections at once by default. If every request opened its own connection, the app would be slow, and under load it would run out.

A **pool** is like a taxi rank. It keeps a few connections (taxis) open. Each query takes a free one, uses it, and puts it back. If all are busy, the next query waits a moment in line.

what `pool.query()` does for you

```
pool.query("SELECT ...")
1. borrow a free connection (or open one, up to 10)
2. send the SQL to PostgreSQL
3. wait for the answer
4. give the connection back (closed if unused for ~10 s)
5. hand you the result: { rows: [...] }
```

Why the `globalThis` trick? In development, Next.js re-runs a file each time you save it. Without the trick, every save would create a *new* pool while the old ones keep their connections open. After a few dozen saves Postgres says `sorry, too many clients already`. Parking the pool on `globalThis`, which survives reloads, means there is always just one. In production files aren't reloaded, so the trick isn't needed there.

IF YOU KNOW MONGOOSE / EXPRESS

This file is your `mongoose.connect(process.env.MONGODB_URI)`. Mongoose also keeps a pool behind the scenes; `pg` just makes it visible.

12 Write the model: `models/todo.ts`

Create a folder `models` and this file. It's the **only** file that contains SQL. Each function does one job:

models/todo.ts	
Code	What it means
<pre>import { pool } from "@/lib/db";</pre>	Bring in the shared pool from <code>lib/db.ts</code> . <code>@/</code> means the project root.
<pre>// The "model": everything that talks to the // todos table lives here. // (Like a Mongoose model, but the table's // shape is defined in schema.sql.)</pre>	Comments: this file is the only place that writes SQL. Every other file just calls these functions.

<pre>// Shape of one row in the todos table export type Todo = { id: number; text: string; done: boolean; };</pre>	A TypeScript type describing one row of the table. It matches <code>schema.sql</code> column by column. Other files import it so TypeScript knows what a todo looks like.
<pre>// SELECT = read rows export async function findAllTodos(): Promise<Todo[]> {</pre>	<code>findAllTodos()</code> : returns every todo. <code>async</code> = it takes time (talking to the database). <code>Promise<Todo[]></code> = "later, you'll get an array of Todos".
<pre> const { rows } = await pool.query<Todo>("SELECT * FROM todos ORDER BY id");</pre>	Send SQL to Postgres and <code>await</code> (wait for) the answer. <code><Todo></code> tells TypeScript what each row looks like. The result object has a <code>rows</code> array; <code>{ rows }</code> pulls it out. <code>ORDER BY id</code> = oldest first.
<pre> return rows; }</pre>	Give the array back to whoever called this function.
<pre>// INSERT = add a row. \$1 is a safe placeholder for the value. export async function createTodo(text: string): Promise<Todo> {</pre>	<code>createTodo(text)</code> : adds one row and returns it.
<pre> const { rows } = await pool.query<Todo>("INSERT INTO todos (text) VALUES (\$1) RETURNING *", [text]);</pre>	<code>\$1</code> is a placeholder; the real value comes from the array <code>[text]</code>. Postgres keeps value and SQL separate, so a user typing SQL can't break your query (this is called preventing SQL injection). <code>RETURNING *</code> = send back the row you just made, including its new <code>id</code>.
<pre> return rows[0]; }</pre>	There's exactly one new row, so return the first item.
<pre>// UPDATE = change a row. Returns undefined if no row has that id. export async function updateTodoDone(id: number, done: boolean): Promise<Todo undefined> {</pre>	<code>updateTodoDone(id, done)</code> : changes the done flag of one row.
<pre> const { rows } = await pool.query<Todo>("UPDATE todos SET done = \$1 WHERE id = \$2 RETURNING *", [done, id]);</pre>	<code>\$1</code> gets <code>done</code>, <code>\$2</code> gets <code>id</code> (same order as the array). <code>WHERE id = \$2</code> = only touch that one row.
<pre> return rows[0]; }</pre>	Return the updated row. If no row had that id, <code>rows</code> is empty, so this returns <code>undefined</code> (the API turns that into a 404).
<pre>// DELETE = remove a row export async function deleteTodo(id: number): Promise<void> {</pre>	<code>deleteTodo(id)</code> : removes one row. <code>Promise<void></code> = it returns nothing.
<pre> await pool.query("DELETE FROM todos WHERE id = \$1", [id]); }</pre>	Run the DELETE and wait for it to finish.

IF YOU KNOW MONGOOSE / EXPRESS

This is your `Todo` model. Same jobs, different words:

Mongoose, for comparison only

```
// models/Todo.js (Mongoose)
const todoSchema = new Schema({
  text: { type: String, required: true },
  done: { type: Boolean, default: false },
});
const Todo = model("Todo", todoSchema);

Todo.find().sort({ _id: 1 });
Todo.create({ text });
Todo.findByIdAndUpdate(id, { done }, { new: true });
Todo.findByIdAndDelete(id);
```

PostgreSQL (this project)

```
-- schema.sql (run once in psql)
CREATE TABLE todos (
  id SERIAL PRIMARY KEY,
  text TEXT NOT NULL,
  done BOOLEAN NOT NULL DEFAULT FALSE
);

SELECT * FROM todos ORDER BY id
INSERT INTO todos (text) VALUES ($1) RETURNING *
UPDATE todos SET done = $1 WHERE id = $2
RETURNING *
DELETE FROM todos WHERE id = $1
```

The big difference: Mongoose's schema lives in your **code**. A Postgres schema lives **inside the database** (you made it in step 5), and the database itself rejects bad data. Our `Todo` type just mirrors it for TypeScript.

13 Write the API routes (your controllers)

IN PLAIN ENGLISH

An **API route** is a URL that returns data instead of a page. In Next.js you create one by making a file called `route.ts`; its **folder path** becomes the URL, and each exported function named after an HTTP method handles that method. No router setup and no `app.listen()`: `npm run dev` already serves it.

Our four endpoints:

Method + URL	Send (body)	Get back	File
GET /api/todos	nothing	200 + list of todos	app/api/todos/route.ts
POST /api/todos	<code>{"text": "Buy milk"}</code>	201 + the new todo (or 400)	
PATCH /api/todos/7	<code>{"done": true}</code>	200 + updated todo (or 400/404)	app/api/todos/[id]/route.ts
DELETE /api/todos/7	nothing	204, empty	

Each handler does three things: **read** the request, **check** it, **call the model** and reply. Create the folders `app/api/todos/` and `app/api/todos/[id]/` (yes, with square brackets in the folder name).

app/api/todos/route.ts

Code

```
import { NextResponse } from "next/server";
```

What it means

`NextResponse` is how you build the reply: JSON data plus a status code.

import { createTodo, findAllTodos } from "@models/todo";	Bring in the two model functions this file needs.
// GET /api/todos -> list all todos export async function GET() {	Because this function is named GET , Next.js runs it for GET /api/todos . The URL comes from the folder path app/api/todos/ .
const todos = await findAllTodos();	Ask the model for all todos (it runs the SELECT).
return NextResponse.json(todos); }	Send them to the browser as JSON. Status is 200 (OK) by default.
// POST /api/todos body: { text } -> create a todo export async function POST(req: Request) {	Runs for POST /api/todos . req is the incoming request.
const { text } = await req.json();	Read the JSON the browser sent, e.g. { "text": "Buy milk" } , and take out text .
if (typeof text !== "string" !text.trim()) { return NextResponse.json({ error: "text is required" }, { status: 400 }); }	Validate before touching the database: if text isn't a string, or is only spaces, reply 400 Bad Request and stop.
const todo = await createTodo(text.trim());	Ask the model to INSERT it (with spaces trimmed off the ends).
return NextResponse.json(todo, { status: 201 }); }	Reply with the new todo and status 201 Created .

app/api/todos/[id]/route.ts	
Code	What it means
import { NextResponse } from "next/server";	Same reply helper.
import { deleteTodo, updateTodoDone } from "@models/todo";	The two model functions this file needs.
type Ctx = { params: Promise<{ id: string }> };	Describes the 2nd thing Next.js passes in: params , which holds the [id] part of the URL. For /api/todos/7 , id is "7" (always a string). It's a Promise, so you await it.
// PATCH /api/todos/:id body: { done } -> update a todo export async function PATCH(req: Request, { params }: Ctx) {	Runs for PATCH /api/todos/7 . PATCH = "change part of something".
const { id } = await params;	Wait for params and take out id .
const { done } = await req.json();	Read done from the JSON body, e.g. { "done": true } .
if (typeof done !== "boolean") { return NextResponse.json({ error: "done must be true or false" }, { status: 400 }); }	Validate: must be exactly true or false , otherwise 400 .
const todo = await updateTodoDone(Number(id), done);	Ask the model to UPDATE. Number(id) turns the text "7" into the number 7 .
if (!todo) { return NextResponse.json({ error: "not found" }, { status: 404 }); }	No row had that id: reply 404 Not Found .

<pre> return NextResponse.json(todo); } </pre>	Success: send back the updated todo.
<pre>// DELETE /api/todos/:id -> remove a todo export async function DELETE(_req: Request, { params }: Ctx) { </pre>	Runs for <code>DELETE /api/todos/7</code> . <code>_req</code> starts with <code>_</code> to show we don't use it (we still need it, because <code>params</code> is always the 2nd argument).
<pre> const { id } = await params; </pre>	Get the id from the URL.
<pre> await deleteTodo(Number(id)); </pre>	Ask the model to DELETE that row.
<pre> return new NextResponse(null, { status: 204 }); }</pre>	Reply 204 No Content : "done, nothing to send back".

Try the API before building the page

With `npm run dev` running, open **`http://localhost:3000/api/todos`** in the browser: you'll see `[]`, the JSON for an empty list. To send other methods, use PowerShell:

PowerShell

```
Invoke-RestMethod -Method Post -Uri http://localhost:3000/api/todos -ContentType "application/json"
-Body '{"text":"Buy milk"}'
Invoke-RestMethod http://localhost:3000/api/todos
Invoke-RestMethod -Method Patch -Uri http://localhost:3000/api/todos/1 -ContentType
"application/json" -Body '{"done":true}'
Invoke-RestMethod -Method Delete -Uri http://localhost:3000/api/todos/1
```

TIP

Use `Invoke-RestMethod` in PowerShell instead of `curl`: in Windows PowerShell, `curl` is a different command and the JSON quotes get mangled.

14 Write the Zustand store: `store/todoStore.ts`

IN PLAIN ENGLISH

A **store** is a shared whiteboard for the browser side of the app. It holds the todo list and the actions that change it. Any component can read it with `useTodoStore()`, and when the list changes, every component using it redraws.

With one page you *could* keep everything in `useState`. The store pays off when a second component (say, a counter in a header) needs the same list: no passing props down through layers. It also moves the `fetch` calls out of the page, so the page only handles display.

`store/todoStore.ts`

Code	What it means
<code>import { create } from "zustand";</code>	<code>create</code> is Zustand's one main function. It builds a store.
<code>import type { Todo } from "@models/todo";</code>	Reuse the <code>Todo</code> type from the model. <code>import type</code> brings in only the type, never the database code, so it's safe in the browser.

<pre>// What the store holds: data (todos) + actions (functions that change it) type TodoStore = { todos: Todo[]; loadTodos: () => Promise<void>; addTodo: (text: string) => Promise<void>; toggleTodo: (todo: Todo) => Promise<void>; removeTodo: (id: number) => Promise<void>; };</pre>	The TypeScript type of the store: one piece of data (<code>todos</code>) and four actions (functions). <code>() => Promise<void></code> = "a function that does async work and returns nothing".
<pre>// create() makes a hook. Any component can call useTodoStore() to read or change it. // set = replace part of the state; get = read the current state.</pre>	Comments on the next line.
<pre>export const useTodoStore = create<TodoStore> ((set, get) => ({</pre>	Build the store. You get two helpers: <code>set</code> (change the data, which re-draws any component using it) and <code>get</code> (read the current data). The result, <code>useTodoStore</code>, is a hook that components call.
<pre> todos: [],</pre>	Starting data: an empty list.
<pre> loadTodos: async () => { const res = await fetch("/api/todos"); set({ todos: await res.json() }); },</pre>	Action: GET the list from the API, then <code>set({ todos: ... })</code>. Every component showing <code>todos</code> re-draws automatically.
<pre> addTodo: async (text) => {</pre>	Action: add a todo.
<pre> await fetch("/api/todos", { method: "POST", headers: { "Content-Type": "application/json" }, body: JSON.stringify({ text }), });</pre>	Send a POST request. <code>headers</code> says "the body is JSON"; <code>JSON.stringify</code> turns <code>{ text }</code> into a JSON string.
<pre> await get().loadTodos(); },</pre>	Then reload the list from the database, so the screen shows exactly what's saved.
<pre> toggleTodo: async (todo) => {</pre>	Action: flip done/not done for one todo.
<pre> await fetch(`/api/todos/\${todo.id}`, { method: "PATCH", headers: { "Content-Type": "application/json" }, body: JSON.stringify({ done: !todo.done }), });</pre>	Send a PATCH to <code>/api/todos/<id></code>. The backticks <code>`...\${}`</code> insert the id into the URL. <code>!todo.done</code> = the opposite of its current value.
<pre> await get().loadTodos(); },</pre>	Reload the list.
<pre> removeTodo: async (id) => {</pre>	Action: delete one todo.
<pre> await fetch(`/api/todos/\${id}`, { method: "DELETE" });</pre>	Send a DELETE to <code>/api/todos/<id></code>.
<pre> await get().loadTodos(); }, });</pre>	Reload the list. The <code>});</code> closes the store object, the arrow function and <code>create(</code>.

15 Replace app/layout.tsx

Replace the generated file with this simpler version. (The generated one uses `LayoutProps<"/">`, a newer Next.js helper type; both work.)

app/layout.tsx	
Code	What it means
<pre>// layout.tsx wraps every page. It must render <html> and <body>.</pre>	Comment: the layout is the frame around every page.
<pre>export const metadata = { title: "To-Do" };</pre>	Page info for the browser. <code>title</code> is the text on the browser tab.
<pre>export default function RootLayout({ children }: { children: React.ReactNode }) {</pre>	The layout component. Next.js gives it <code>children</code> (the page being shown). The type says <code>children</code> is anything React can display.
<pre> return (</pre>	What this component draws.
<pre> <html lang="en"></pre>	The outer HTML tag. Next.js requires the root layout to include <code><html></code> and <code><body></code> . <code>lang="en"</code> = English page.
<pre> <body style={{ fontFamily: "sans-serif", maxWidth: 400, margin: "40px auto" }}></pre>	The page body. The <code>style</code> uses a simple font, max width 400px, and <code>margin: "40px auto"</code> centers it.
<pre> {children}</pre>	The current page (<code>page.tsx</code>) is placed here.
<pre> </body> </html></pre>	Closing tags.
<pre>); }</pre>	End of what's drawn, end of the function.

16 Replace app/page.tsx

The page only **displays** things and **calls store actions**. It never talks to the API or the database directly.

app/page.tsx	
Code	What it means
<pre>"use client"; // runs in the browser: needed for hooks and click handlers</pre>	This component runs in the browser . Needed because it uses hooks and click handlers. (Without it, Next.js runs components on the server.)
<pre>import { useEffect, useState } from "react";</pre>	Two built-in React hooks: <code>useState</code> (remember a value) and <code>useEffect</code> (run code after the page shows).
<pre>import { useTodoStore } from "@/store/todoStore";</pre>	Our Zustand store hook.
<pre>export default function Home() {</pre>	The page. Next.js shows it at <code>/</code> because it's <code>app/page.tsx</code> .
<pre> // Shared state + actions come from the Zustand store const { todos, loadTodos, addTodo, toggleTodo, removeTodo } = useTodoStore();</pre>	Grab the todo list and the four actions from the store. When the store's <code>todos</code> change, this page re-draws.

<pre>// Local state: only this page cares what's typed in the box const [text, setText] = useState("");</pre>	<code>text</code> = what's typed in the box. Only this page needs it, so plain <code>useState</code> is enough. Rule of thumb: used by one component = <code>useState</code> ; shared data = store.
<pre>// Load todos from the database once, when the page opens useEffect(() => { loadTodos(); }, [loadTodos]);</pre>	After the page first appears, call <code>loadTodos()</code> . The <code>[loadTodos]</code> list means "run again only if this changes", and it never does, so it runs once.
<pre>async function handleAdd() {</pre>	Runs when you click Add or press Enter.
<pre> if (!text.trim()) return;</pre>	Ignore empty input.
<pre> await addTodo(text);</pre>	Call the store action (which POSTs to the API and reloads the list).
<pre> setText(""); }</pre>	Clear the input box.
<pre>return (</pre>	Below is JSX: HTML-like code that describes the screen.
<pre> <main> <h1>To-Do</h1></pre>	A container and a heading.
<pre> <input value={text} onChange={(e) => setText(e.target.value)} onKeyDown={(e) => e.key === "Enter" && handleAdd()} placeholder="New task..." /></pre>	The text box. <code>value={text}</code> shows the state. <code>onChange</code> saves each keystroke into the state. <code>onKeyDown</code> : Enter calls <code>handleAdd()</code> . <code>placeholder</code> = grey hint text.
<pre> <button onClick={handleAdd}>Add</button></pre>	The Add button.
<pre> </pre>	A bullet list.
<pre> {todos.map((t) => (</pre>	<code>{ }</code> = "JavaScript here". <code>.map</code> turns each todo into one <code></code> .
<pre> <li key={t.id}></pre>	One item. <code>key</code> = a unique id so React can track items. Use the database id.
<pre> <input type="checkbox" checked= {t.done} onChange={() => toggleTodo(t)} /></pre>	Shows <code>done</code> . Clicking calls the store's <code>toggleTodo</code> .
<pre> {t.text}</pre>	The task text, crossed out if done.
<pre> <button onClick={() => removeTodo(t.id)}>x</button></pre>	Delete button. <code>() => ...</code> wraps the call so it runs on click, not immediately.
<pre>)}}</pre>	End of one item; end of <code>.map</code> .
<pre> </main>); }</pre>	Close the list and container; end of the component.

17 Run and test

```
PowerShell
```

```
npm run dev
```

Open **http://localhost:3000**, then check:

1. Type a task and press Enter. It appears and the box clears.
2. Tick it. It's crossed out.
3. **Refresh the page.** It's still there, because it came back from PostgreSQL.
4. Click **x**. It's gone.
5. Look in the database directly: `psql -U postgres -h localhost -d todo_app -c "SELECT * FROM todos;"`

Part 5: Express + Mongoose vs Next.js + PostgreSQL

Same ideas, different places.

Job	Express + Mongoose	This project
Start the server	<code>app.listen(3000)</code> in <code>server.js</code>	<code>npm run dev</code> (Next.js is the server)
Connect to the database	<code>mongoose.connect(uri)</code>	<code>new Pool({ connectionString })</code> in <code>lib/db.ts</code>
Define the data shape	<code>new Schema({...})</code> in code	<code>CREATE TABLE</code> in <code>schema.sql</code> , run once + <code>Todo</code> type
Model / queries	<code>Todo.find()</code> , <code>Todo.create()</code> ...	Functions in <code>models/todo.ts</code> running SQL
Routes	<code>router.get("/todos", ctrl.list)</code>	Folder <code>app/api/todos/</code> + exported <code>GET</code>
Controllers	<code>(req, res) => res.json(...)</code>	<code>GET(req)</code> returning <code>NextResponse.json(...)</code>
URL parameter	<code>req.params.id</code>	<code>[id]</code> folder + <code>await params</code>
Request body	<code>req.body</code> (needs <code>express.json()</code>)	<code>await req.json()</code>
IDs	<code>_id : ObjectId string</code>	<code>id : 1, 2, 3...</code> (SERIAL)
Shell	<code>mongosh</code>	<code>psql</code>
GUI	MongoDB Compass	pgAdmin 4
Frontend	separate React app + CORS setup	same project, same address; no CORS

Express + Mongoose, for comparison only

```
// Express: routes + controller
router.post("/todos", async (req, res) => {
  const { text } = req.body;
  if (!text?.trim())
    return res.status(400).json({ error: "text is required" });
  const todo = await Todo.create({ text });
  res.status(201).json(todo);
});
```

this project (shortened)

```
// Next.js: app/api/todos/route.ts
export async function POST(req: Request) {
  const { text } = await req.json();
  if (typeof text !== "string" || !text.trim())
    return NextResponse.json({ error: "text is required" }, { status: 400 });
  const todo = await createTodo(text.trim());
  return NextResponse.json(todo, { status: 201 });
}
```

IN PLAIN ENGLISH

Do I need Express? No. Next.js route handlers already do Express's job (routing, reading requests, sending JSON), and they run on Node.js. You'd add Express only if you wanted a separate backend server.

Want Mongoose-style models with Postgres? That's what an **ORM** is. **Prisma** and **Drizzle** are the popular ones: you describe tables in code and write `db.todo.findMany()` instead of SQL. Learn raw SQL first (this project); ORMs make more sense once you know what they generate.

Part 6: Memorize and practice

The journey of one click

What happens when you type "Buy milk" and press Enter:

1. `page.tsx`: `onKeyDown` sees Enter and calls `handleAdd()`.
2. `handleAdd()` calls the store action `addTodo("Buy milk")`.
3. `todoStore.ts`: `fetch("/api/todos", { method: "POST", body: '{"text":"Buy milk"}' })`.
4. The request crosses from browser to server. `Next.js` finds `app/api/todos/route.ts` and runs `POST`.
5. `POST` reads the JSON, checks `text`, calls `createTodo("Buy milk")`.
6. `models/todo.ts`: `pool.query("INSERT ... VALUES ($1) RETURNING *", ["Buy milk"])`.
7. The pool lends a connection; PostgreSQL saves the row and returns `{id: 1, text: "Buy milk", done: false}`.
8. `POST` replies **201** with that JSON.
9. Back in the store: `loadTodos()` does a GET and calls `set({ todos })`.
10. React sees the store changed and redraws the list. "Buy milk" appears.

One table to rule them all

Action	Page	Store action	HTTP	Route function	Model function	SQL
Read	<code>useEffect</code>	<code>loadTodos</code>	GET <code>/api/todos</code>	GET	<code>findAllTodos</code>	SELECT
Create	<code>handleAdd</code>	<code>addTodo</code>	POST <code>/api/todos</code>	POST	<code>createTodo</code>	INSERT
Update	checkbox	<code>toggleTodo</code>	PATCH <code>/api/todos/:id</code>	PATCH	<code>updateTodoDone</code>	UPDATE
Delete	x button	<code>removeTodo</code>	DELETE <code>/api/todos/:id</code>	DELETE	<code>deleteTodo</code>	DELETE

Rules to remember

- `app/` folder path = URL. `page.tsx` = page. `route.ts` = API. `[id]` = changing part.
- Exported function name in `route.ts` = HTTP method.
- `"use client"` = runs in the browser. Needed for hooks and clicks. Database code must **never** be in a client file.
- Only the model writes SQL. Routes check input and call the model. The page only calls the store.
- SQL values always go in `$1`, `$2` placeholders, never glued into the string.
- Secrets go in `.env.local`, read as `process.env.NAME`, on the server only. Restart after changing.
- One component needs it: `useState`. Many components share it: the store.
- One pool for the whole app; `pool.query()` borrows and returns connections for you.

Practice ladder (do them in order)

1. **Rebuild from memory.** Delete `lib`, `models`, `store`, `app/api` and `page.tsx`, then rebuild using only the table above.
2. **Count.** Show "2 of 5 done" above the list. Compute it from `todos` in the page (no new state).

3. **Filter (Zustand).** Add `filter: "all" | "active" | "done"` and a `setFilter` action to the store; add three buttons. Browser-only, no API change.
4. **Created date (SQL).** In `psql`: `ALTER TABLE todos ADD COLUMN created_at TIMESTAMP NOT NULL DEFAULT now();` then add it to the `Todo` type and show it.
5. **Edit text (full stack).** Model `updateTodoText`, accept `text` in `PATCH`, a store action, and an Edit button.
6. **Clear completed.** `DELETE FROM todos WHERE done = true`: a model function, a `DELETE /api/todos` handler, a store action and a button.
7. **Errors and loading.** Add `loading` and `error` to the store; check `res.ok` after each fetch; show "Loading..." and error messages.
8. **Optimistic update.** In `toggleTodo`, flip it in the store *first*, then call the API; undo if it fails.
9. **Try an ORM.** Rebuild `models/todo.ts` with Prisma or Drizzle and compare.

Troubleshooting

You see	Fix
<code>psql</code> is not recognized	Add <code>C:\Program Files\PostgreSQL\17\bin</code> to PATH (step 3) and open a new terminal.
password authentication failed for user "postgres"	The password in <code>.env.local</code> (or what you typed) is wrong. Restart <code>npm run dev</code> after fixing.
<code>ECONNREFUSED 127.0.0.1:5432</code>	PostgreSQL isn't running. Run <code>sc.exe query postgresql-x64-17</code> ; start it in the Services app.
database "todo_app" does not exist	Do step 4.
relation "todos" does not exist	Do step 5, and make sure you used <code>-d todo_app</code> .
sorry, too many clients already	Something creates many pools. Make sure only <code>lib/db.ts</code> calls <code>new Pool</code> .
Cannot find module '@/...'	Check the file path and the <code>paths</code> entry in <code>tsconfig.json</code> .
An error saying <code>useState</code> / <code>useEffect</code> only works in a Client Component	Add <code>"use client";</code> as the first line of the page.
Page loads but the list stays empty	Open <code>http://localhost:3000/api/todos</code> . If it shows an error, read the <code>npm run dev</code> terminal; the real error is printed there.

Generated from the project's real source files by `guide/build-guide.mjs`. The `guide/` folder is not part of the app.